OPEN SOURCE APACHE 2.0

MCP Gateway & AI Registry

Unified Control Plane for AI Agents, MCP Servers, and Agent Skills

MCP SERVERS A2A AGENTS AGENT SKILLS ENTERPRISE READY

GitHub Repository AWS Workshop

Last updated: April 11, 2026

The AI Asset Sprawl Problem

Without a Control Plane

- ✗ Separate connections per MCP server
- ✗ Each developer configures tools individually
- ✗ Scattered API keys and credentials
- ✗ No visibility into tool usage
- ✗ No dynamic tool discovery for agents
- ✗ Separate agent and server governance

With Gateway & Registry

- ✓ One gateway, access all MCP servers
- ✓ Single config for VS Code, Cursor, Claude Code
- ✓ Centralized credential management
- ✓ Complete audit trails and analytics
- ✓ Semantic discovery for servers, agents, skills
- ✓ Unified governance for all AI assets

A central control plane for all your AI assets -- servers, agents, and skills

Four Core Functions

Unified MCP Gateway

Centralized access point for multiple MCP servers with reverse proxy routing, SSL termination, and health monitoring.

MCP Server Registry

Register, discover, and manage MCP servers with unified governance, semantic search, and version routing.

Agent Registry & A2A Hub

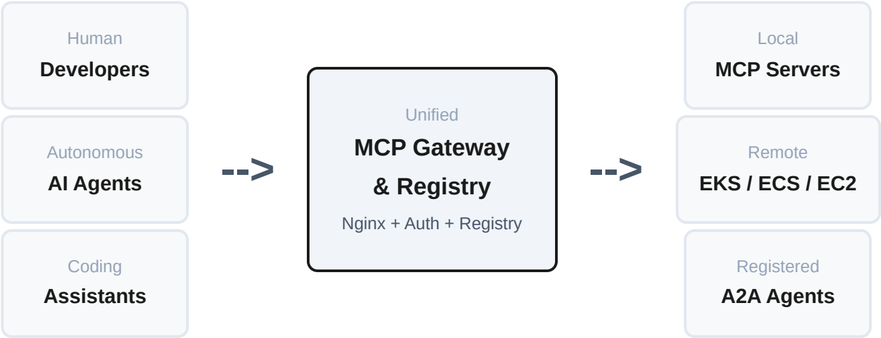
Agent registration, discovery, and direct agent-to-agent communication via the A2A Protocol. Supports both A2A and non-A2A agents.

Skills Registry

Register, version, and discover reusable AI skills (SKILL.md) with security scanning, enabling skill sharing across agents and teams.

Platform Demo

Platform Architecture



Single connection point with **dual authentication** supporting both human users and machine-to-machine agent flows.

Enterprise Security

Multi-Provider IAM
Keycloak, Microsoft Entra ID, Okta, Auth0, and AWS Cognito with harmonized management API.

Fine-Grained Access
Scopes define which servers, methods, tools, and agents each user or service account can access.

Security Scanning
Cisco AI Defense integration for MCP servers and A2A agents with YARA pattern matching.

Audit Logging
Comprehensive audit trails with credential masking, TTL retention, and SOC 2/GDPR compliance.

M2M Auth
OAuth2 Client Credentials flow for AI agent identity with service accounts.

Static Token Auth
API key access for CI/CD pipelines and trusted network environments without IdP setup.

Intelligent Discovery

Hybrid Search

- Vector similarity + keyword matching
- Unified search across servers, tools, agents, skills
- Name-boost for exact match relevance
- Tag-based and metadata filtering

Flexible Embeddings

- Local sentence-transformers
- OpenAI embeddings
- Amazon Bedrock Titan
- 100+ LiteLLM-supported providers

Agents discover tools at runtime via natural language -- **"find an agent that can book flights"**

Agent Registry & A2A Protocol

Agent-to-Agent Communication

Direct peer-to-peer agent communication. Registry handles discovery, agents connect directly for low-latency interaction.

Semantic Agent Discovery

Agents find collaborators via natural language queries. Dynamic composition based on capabilities, not hardcoded references.

Agent Skills Registry

Register reusable SKILL.md instruction sets with security scanning, visibility controls, and star ratings.

Agents autonomously discover specialized agents for tasks they cannot handle, enabling **dynamic agent orchestration**.

Agent Trust: Agent Name Service

DNS + PKI for AI Agents -- just as SSL certificates let you trust websites, ANS lets you trust agents.

Structured Identity

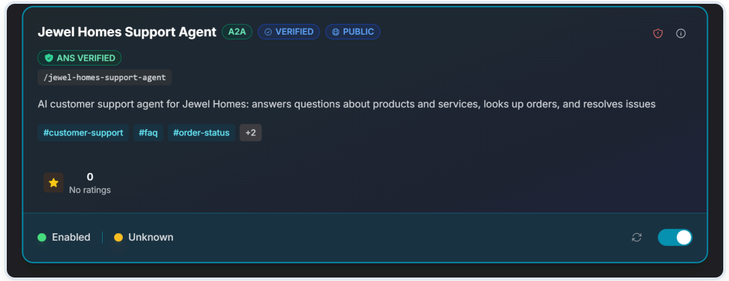
Each agent gets an ANS Name and X.509 certificate issued by a Registration Authority.

Bring Your Own ANS ID

Owners link their ANS Agent ID to registry entries. Registry verifies via GoDaddy ANS API.

Production Resilience

Circuit breaker, retry with backoff, 6-hour background re-verification, and rate limiting.



Verified agents display a trust badge with certificate details

IETF Internet-Draft (draft-narajala-ans-00) -- Powered by GoDaddy ANS, integrated into the open-source AI Registry

Registry Federation

P2P Federation

Bidirectional sync between registry instances with configurable sync modes: all, whitelist, or tag filter.

Anthropic Registry

Import curated MCP servers from Anthropic's official registry with full API compatibility.

External Sources

AWS Agent Registry, Workday ASOR, and Anthropic Registry -- federate assets from multiple external sources.

Registry Card

Standardized discovery via /.well-known/registry-card with capabilities, auth endpoints, and contact info.

Central IT aggregates across LOB registries -- LOBs inherit shared tools from a **central hub**.

AWS Agent Registry Federation

Federate MCP servers, A2A agents, and skills from AWS Agent Registry into the Gateway.

Multi-Registry Support

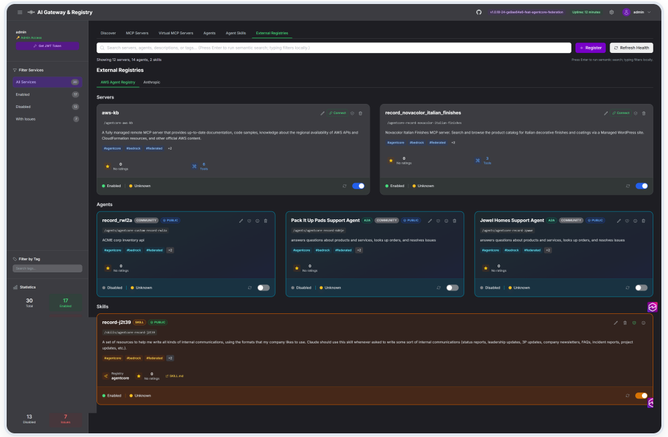
Add multiple registries from same or different AWS accounts and regions. Cross-account via IAM role assumption.

Selective Sync

Choose descriptor types (MCP, A2A, CUSTOM, AGENT_SKILLS) and status filters per registry.

Cascade Cleanup

Remove a registry and all its synced assets are automatically deregistered.



Federated assets from AWS Agent Registry in the External Registries tab

Enable with a single env var: **AWS_REGISTRY_FEDERATION_ENABLED=true** -- manage registries via UI, API, or CLI | **Watch Demo**

Virtual Servers & Version Routing

Virtual MCP Servers

- Aggregate tools from multiple backends
- Tool aliasing for naming conflicts
- Per-tool scope-based access control
- Session multiplexing (1:N transparent)
- 60s cached aggregation

Version Routing

- Multiple versions behind single endpoint
- Test with X-MCP-Server-Version header
- Instant rollback with one API call
- Deprecation lifecycle with sunset dates
- O(1) nginx map-based routing

Observability & Metrics

Grafana Dashboards

Real-time metrics for server health, tool usage, auth events, and system performance.

OTLP Push Export

Direct export to Datadog, New Relic, Honeycomb, or Grafana Cloud -- no OTEL Collector needed.

OpenTelemetry

Integration with Prometheus, CloudWatch, and other monitoring platforms via OTEL protocol.

Anonymous Telemetry

Privacy-first usage telemetry for adoption tracking. Non-sensitive metadata only. Opt-out with one env var.

Flexible Deployment

Amazon ECS Fargate

Serverless containers with multi-AZ, ALB with HTTPS, auto-scaling, CloudWatch, and NAT Gateway HA. Full Terraform config.

Amazon EKS (Kubernetes)

Helm chart deployment on Amazon EKS for container orchestration at scale.

Amazon EC2

Traditional deployment on EC2 instances with Docker Compose. Complete setup guide from scratch.

Docker / Podman

Pre-built Docker Hub images and rootless Podman support for macOS and Linux. One-command AI-assisted setup.

3
AWS Deployment Targets

5
Identity Providers

700+
Passing Tests

Developer Experience

MCP Registry CLI

Conversational interface for registry management with real-time token status and cost tracking.

IAM Settings UI

Visual management of users, groups, and M2M service accounts with fine-grained permissions.

AI Coding Assistants

Single config for VS Code, Cursor, Claude Code, and Cline with guided tool discovery.

Management API

RESTful HTTP endpoints with Python client, type-safe interfaces, and comprehensive error handling.

Config Viewer

Admin panel with 11 config groups. Export as ENV, JSON, TFVARS, or YAML for deployment automation.

Custom Metadata

Rich metadata on servers and agents for team ownership, compliance tracking, and cost allocation.

What's New

- **AWS Agent Registry Federation**
  Federate MCP servers, A2A agents, and skills from AWS Agent Registry. Multi-registry, cross-account, cascade cleanup.
- **Register Any Agent (A2A and Non-A2A)**
  New supported_protocol field distinguishes agent types. Register through UI, API, or CLI.
- **Amazon Bedrock AgentCore Bulk Import**
  Auto-discover and register all AgentCore Gateways and Agent Runtimes from your AWS account.
- **Agent Name Service (ANS) Integration**
  PKI-based trust verification for agents via GoDaddy ANS. Green shield badge on verified agents.
- **Auth0 Identity Provider Support**
  Fifth IdP alongside Keycloak, Entra ID, Okta, and Cognito. Harmonized IAM API for all providers.
- **Anonymous Usage Telemetry**
  Privacy-first deployment metadata. No PII. Opt-out with MCP_TELEMETRY_DISABLED=1.
- **Registry Card for Federation Discovery**
  Standardized /.well-known/registry-card with capabilities and auth metadata.
- **AWS Workshop Studio**
  Hands-on workshop for deploying and securing AI agent ecosystems on AWS.
- **AI Registry MCP Server (airegistry-tools)**
  5 tools for AI coding assistants to discover and query registry directly.

What's Next

Registry Copilot An AI assistant that helps you build, register, and configure your agents through natural language

Enterprise Readiness Enhanced UI dashboards, structured audit logging, RBAC refinements, and advanced security controls

Have a feature idea?

We build in the open. Tell us what you need.

Create an issue on GitHub →

Demo & Links

AWS Show & Tell

YouTube -- MCP Gateway & Registry deep dive

Live Demo

Vidcast -- Hands-on walkthrough of the platform

Roadmap

GitHub Milestones -- Upcoming features and releases

One AI Registry. Unified Control.

3
AWS Targets
ECS Fargate, EKS, EC2

5
Identity Providers
Keycloak, Entra ID, Okta, Auth0, Cognito

Open
Source
Apache 2.0 License

github.com/agentic-community/mcp-gateway-registry

Quick Start | **Documentation** | **AWS Deployment**

AWS Workshop | **Demo Video** | **Community**